

in the margin of a copy of Diophantus' *Arithmetica*, which was later known as "Fermat's last theorem". The theorem states, quite simply, that it is impossible to have  $a^n + b^n = c^n$  for any value of  $n$  larger than 2. Thus, we know that 3 squared + 4 squared = 5 squared (which is basically the relation of the sides (measuring 3 and 4 units) of a right-angled triangle with its hypotenuse (measuring 5 units). In the margin he wrote the theorem, and added that he could prove it, but there wasn't enough space in the margin to do so. This, we believe, is the troll in Fermat rearing its ugly head, and tempting him to torture mathematicians for over 300 years!



This little sculpture was made in a shed, but still managed to fool art experts, and was sold for 440,000 pounds!

Although possible, because people from the 17<sup>th</sup> century seem to be pretty brilliant (Newton certainly was), we feel that Fermat was lying, deliberately. Perhaps the answer had frustrated him, and he decided to play a little mischief by leaving it as a mystery for thousands after him to try and solve. Over 350 years later, in 1995, British mathematician Andrew John Wiles (now knighted), finally cracked the problem using complex modern mathematical techniques that only a handful of people on Earth can really understand, in

a proof that takes up about 150 pages of a book, and took Wiles seven years to come up with! Fermat wasn't lying about one thing: 150 pages of modern math would certainly not fit in the margin of an old French text book!

We don't need to stick to classic history, we can also look at ancient mythology. Whether you believe mythology to be true events or man-made stories doesn't really matter to the context being provided here, because either way you will have to accept that the stories from mythology really do affect the humans of ancient times. Thus we have young Krishna, the naughty prankster, who is much loved by everyone, despite his naughtiness, because he does make people laugh. Norse mythology also contains a deity in the form of Loki, who is a shape-shifter and often fools even the gods in mythology with his powers of deception. History later made him mischievous, but he started out just deceptive.

We can assume, however, that there has always been a desire to be naughty, and fool people. As we mentioned earlier, it doesn't matter what you believe, because even if you believe that naughty deities exist, it still shows that "trolling" really has been around for a long, long time.

But is it a purely human trait?

## IS THERE A TROLL GENE?

As we said at the start of this story, there is no way for us to ever have even a smidgeon of real genetics and evolutionary biology knowledge to allow us to comment on the subject, so please treat this as a wild thought experiment based only on conjecture, and not actual scientific fact.

In our thought experiment, given that genetics and epigenetics shows us new aspects of life that genes can control, and seemingly a lot of mental structure certainly seems genetic (or epigenetic), we feel it's plausible that even things such as comedy and trolling might be genetic in nature. There are just people we all know who are inherently funny, and have a comic timing and way of thinking that most of us can never even hope to emulate. Even with months or years of preparation, we'd never be able to make people laugh

the way these individuals can using only their wit. Thus, there's fundamentally something different in the way some people think. Perhaps we can go as far as to say that there are ways in which their brain just develops very differently.

There are many genetic markers that scientists are associating with differences or deficiencies in the brains of various people. It's why some people have autism, and as a result bad social skills, but can also have wild talents for some area that normal people just cannot fathom. Many claim that Einstein and Newton were probably autistic. What this means is that genes and gene expressions can affect the way our brains develop and function, and perhaps, just perhaps, could even control personality traits such as the ones trolls exhibit. If this is true, a troll might have no control over the desire to troll, and thus could be considered as helpless as you are about your skin colour – which, unless you're Michael Jackson, is something you will have to learn to accept.

## DO ANIMALS TROLL?

There's another way to test for this thought experiment, and we should specify that it's not a truly scientific method as no actual genes are being sampled. If comedy, mischief and trolling really has a genetic origin, then it stands to reason that humans shouldn't be the only ones exhibiting this behaviour.

There are many examples in nature, and the most obvious one is the Cuckoo, who lays eggs in another species nest, so that all the headache of chick rearing and feeding is done by someone else. The human equivalent is having babies, being capable of caring for them, but leaving them on the doorsteps of good samaritans, so that you never have to take the trouble. Some species of cuckoo only breed in this way, while others are known to lay eggs in nests of others of their own species. What this amounts to, in essence, is a spreading of one's genetic offspring into many different nests, and thus increasing the chances of genetic survival. The fact that it's genetic is exhibited by the young bird when it hatches, as it hatches faster